



# Artificial Intelligence-Driven Plant Disease Detection and Diagnosis: A Comprehensive Review of Deep Learning Approaches, Multimodal Sensing Technologies, and Future Perspectives in Precision Agriculture

Surakshya Ghimire<sup>1</sup>, Rajan Lamsal <sup>2\*</sup>, Anvesh Sankuratri<sup>2</sup>, Pradeep Lakkarsu<sup>1</sup>, and Sairam Vutla<sup>3</sup>

<sup>1</sup>Department of Biotechnology (Master of Science), Fort Valley State University, Fort Valley GA 31030, USA

<sup>2</sup>Department of Agricultural Science & Engineering, Tennessee State University, Nashville, TN 37209, USA

<sup>3</sup>Department of Molecular Genetics, Plant Breeding & Biotechnology, University of Florida, Gainesville, FL 32611, USA

(Received on January 9, 2026; Revised on February 16, 2026; Accepted on February 22, 2026)

Plant diseases remain a major threat to global food production, causing significant yield losses and economic impact worldwide. Early and precise disease detection is crucial for effective crop management, yet conventional diagnostic approaches are often slow, labor-intensive, and rely on specialized expertise that may not be widely accessible. Recent advances in artificial intelligence (AI), particularly deep learning-based image analysis, offer scalable and automated solutions for plant disease recognition. This review critically examines forty-one peer-reviewed studies published between 2008 and 2025, selected following PRISMA guidelines from major scientific databases. We summarize key methodological developments, including convolutional neural networks, vision transformers, transfer and few-shot learning, and multimodal sensing approaches, highlighting their reported performance and limitations. Although many models achieve high accuracy in controlled datasets, their effectiveness often decreases under real-field conditions due to environmental variability, limited training data, and practical

deployment constraints. We discuss existing challenges and propose future research directions, emphasizing improved robustness in field environments, development of lightweight and explainable models suitable for edge deployment, and integration with precision agriculture systems. This review aims to guide the design of reliable, practical, and scalable AI-driven plant disease detection strategies.

**Keywords :** CNN, deep learning, multimodal sensing, plant disease detection, precision agriculture

\*Corresponding author.

Phone) +1-478-777-2105, FAX) +1-478-777-2105

E-mail) rajan.plp07@gmail.com

ORCID

Rajan Lamsal

https://orcid.org/0009-0005-9272-0092

Handling Editor : Ki-Tae Kim

© This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (<http://creativecommons.org/licenses/by-nc/4.0>) which permits unrestricted noncommercial use, distribution, and reproduction in any medium, provided the original work is properly cited.

Articles can be freely viewed online at [www.ppjonline.org](http://www.ppjonline.org).

Feeding nearly 8 billion people worldwide makes crop protection more important than ever, and the need for reliable plant disease detection systems has grown accordingly (United Nations Department of Economic and Social Affairs, 2024). Plant pathogens represent a major barrier to achieving the UN's 2030 Zero Hunger goal, often reducing crop yields long before visible symptoms, such as wilting leaves, appear. The Sustainable Development Goals (SDGs), adopted by the United Nations General Assembly in 2015, provide a universal framework for addressing global challenges including poverty, hunger, and environmental sustainability. SDG 2, "Zero Hunger," specifically targets ending hunger, achieving food security, improving nutrition, and promoting sustainable agriculture by 2030 (United Nations General Assembly, 2015). The Food and Agriculture Organization (FAO) estimates that pests and diseases destroy around 40% of global harvests each year, causing economic losses exceeding USD 220 billion. These figures may underestimate the true impact in regions with limited monitoring infrastructure (FAO, 2023). Savary et al. (2019) further reported that pests and pathogens reduce

yields by 21–30% across major crops, including wheat, rice, maize, potato, and soybean, with the most severe effects occurring in food-insecure regions experiencing rapid population growth and recurring disease outbreaks.

Traditionally, plant disease diagnosis relied on experts physically examining crops, a method that works for small plots but is insufficient for large-scale agriculture (Pydipati et al., 2006). Bock et al. (2010) provided a comprehensive overview of visual disease assessment in phytopathometry, documenting that expert visual rating remains the most common method for disease quantification, yet is inherently limited by inter- and intra-rater variability, subjectivity in symptom severity scoring, and difficulty in detecting early or subclinical infections. Nutter et al. (1991) similarly demonstrated that visual disease estimates by human raters are subject to substantial error and can vary significantly even among trained pathologists. This approach suffers from several limitations: scarcity of trained personnel in rural areas, subjective assessments that can vary between observers, high labor costs, and, most importantly, delayed diagnosis after pathogens have already established themselves (Lowe et al., 2017; Mahlein, 2016). These constraints frequently lead to missed detection windows, inappropriate treatment decisions, and ultimately greater yield losses, particularly in resource-limited agricultural systems where access to diagnostic laboratories is restricted (Barbedo, 2018; Singh et al., 2020). Deep learning and computer vision technologies are specifically designed to overcome these limitations by enabling objective, consistent, rapid, and scalable analysis of plant images without the need for on-site expert availability. These challenges have motivated the adoption of artificial intelligence (AI), particularly deep learning and computer vision, for automated and scalable disease detection. Deep learning refers to a class of machine learning algorithms that use multi-layered artificial neural networks to learn hierarchical representations directly from raw data such as images, eliminating the need for manual feature engineering (LeCun et al., 2015). Computer vision, a subfield of AI concerned with enabling machines to interpret visual information, provides the foundational framework for applying these deep learning models to plant disease image analysis.

The deep learning revolution in computer vision began with AlexNet, introduced by Krizhevsky et al. (2012), which achieved a top 5 error rate of 15.3% in the ImageNet Large Scale Visual Recognition Challenge, reducing the previous best error rate by more than 10 percentage points and demonstrating the superiority of deep convolutional networks over traditional hand-crafted feature approaches. Subsequent innovations built on this success: VGGNet

highlighted the benefits of deeper networks, GoogLeNet introduced multi-scale processing, ResNet solved the vanishing gradient problem with skip connections, and EfficientNet demonstrated that intelligent scaling can outperform brute-force depth, enabling application to complex tasks such as plant disease detection.

In agriculture, the release of the PlantVillage dataset by Hughes and Salathé (2015) provided an important early benchmark for deep learning research in plant disease detection. It provides over 54,000 expertly labeled images of healthy and diseased leaves from 14 crops across 38 classes. PlantVillage has become a widely cited resource, showing that smartphone-based diagnosis can make expert knowledge more accessible to farmers. However, it is critical to acknowledge that PlantVillage images were captured under controlled laboratory conditions with uniform, plain backgrounds that do not reflect the complexity of real agricultural environments. Noyan (2022) conducted a rigorous analysis demonstrating that deep learning models trained on PlantVillage frequently exploit background features rather than actual disease symptoms, resulting in significantly degraded performance when deployed on field-collected images. This background bias represents a fundamental limitation that must be carefully considered when interpreting reported accuracy values from this dataset (see Section 5.1 for a detailed discussion). More recently, Vision Transformers (ViTs) have emerged as powerful alternatives to CNNs. Models such as the Swin Transformer have achieved top 1 accuracy of 87.3% on ImageNet-1K (Dosovitskiy, 2020; Liu et al., 2021) and have inspired transformer-based approaches for plant disease detection (Barman et al., 2024; Borhani et al., 2022; Thai et al., 2023).

Despite these advances, applying AI models in real-world field conditions remains challenging. Studies show that models trained under controlled conditions often perform poorly on external images. Mohanty et al. (2016) first reported a dramatic decline in accuracy from 99% to 31% when their CNN model, originally trained on the PlantVillage dataset, was evaluated on field-collected images. Arsenovic et al. (2019) similarly documented substantial accuracy degradation under field conditions and proposed the PlantDiseaseNet dataset, containing 79,265 real-world images, to address these limitations. Bridging this gap is essential for developing reliable and scalable AI-based plant disease detection systems for practical agriculture.

This review summarizes current research on AI-based plant disease detection, tracing developments from traditional machine learning to modern deep learning and vision transformer approaches. Rather than merely cataloging

individual studies, this review offers a critical synthesis by re-evaluating and re-classifying existing technologies through an integrated analytical framework encompassing architectures, learning paradigms, explainability, and deployment constraints. It examines key benchmark datasets, compares model performance, and highlights the main challenges and opportunities for future research. By analyzing 41 peer-reviewed studies published between 2008 and 2025, this review aims to provide researchers, practitioners, and policymakers with a comprehensive and practical understanding of the field.

## Review Methodology

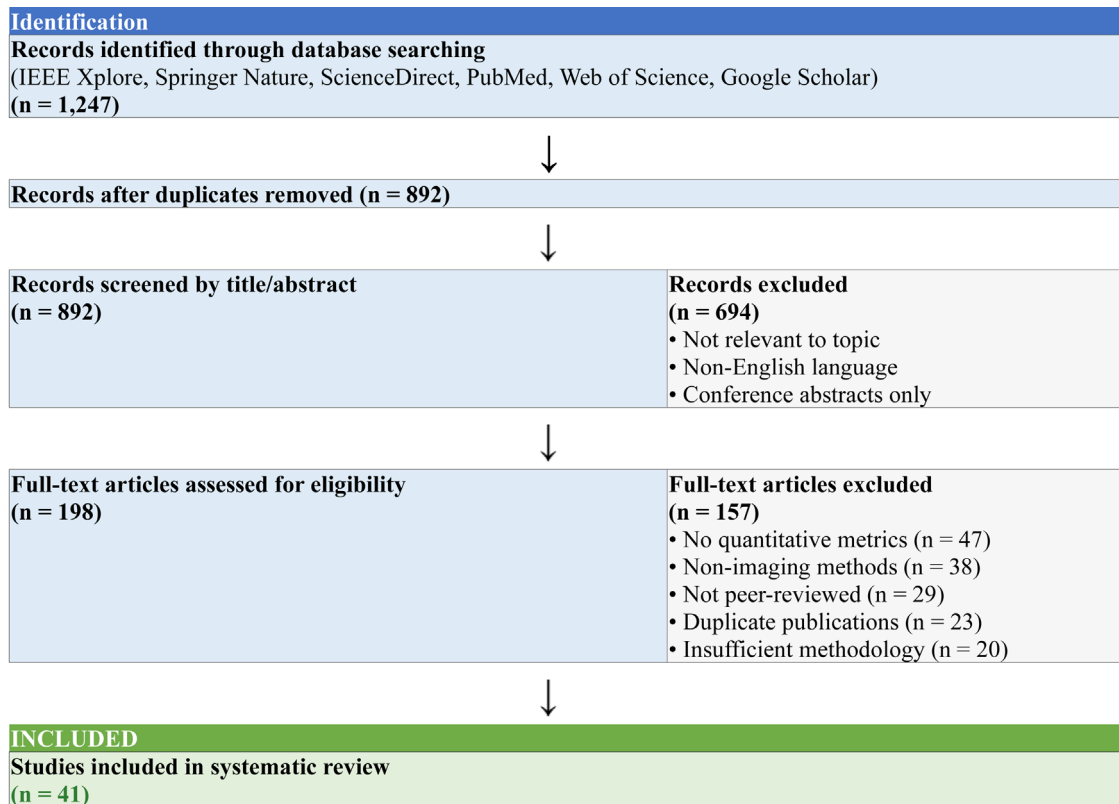
**Literature search strategy.** We performed this systematic review in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Moher et al., 2009). PRISMA is a standardized framework designed to ensure transparency, completeness, and reproducibility in reporting systematic reviews and meta-analyses, and is widely adopted across biomedical and agricultural sciences. Our search spanned six major scientific databases: IEEE Xplore, Springer Nature, Scien-

ceDirect, PubMed, Web of Science, and Google Scholar. The search was conducted between January and March 2025, covering publications from 2008 to early 2025.

The search strategy combined terms related to the disease detection domain (“plant disease detection,” “plant disease classification,” “crop disease identification”) with methodological terms (“deep learning,” “convolutional neural networks,” “CNN,” “vision transformer,” “ViT”) and technique-specific keywords (“transfer learning,” “few-shot learning,” “hyperspectral imaging,” “precision agriculture”). We manually screened reference lists of key review articles to identify additional studies not captured by the database search.

This initial search yielded 1,247 records across all databases. Fig. 1 presents the complete selection process following PRISMA guidelines.

**Data extraction and synthesis.** From each included study, we extracted: bibliographic information (authors, year, journal), target plant species and diseases, dataset characteristics (size, source, image type), deep learning architecture employed, preprocessing and augmentation techniques, experimental design, and reported performance



**Fig. 1.** PRISMA flow diagram illustrates the systematic literature selection process. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

metrics.

Studies were grouped into five methodological categories based on their primary approach: (1) image classification, where models assign disease labels to whole images; (2) object detection, where models localize diseased regions within images; (3) semantic segmentation, where models delineate disease boundaries at pixel level; (4) few-shot and transfer learning approaches designed for limited training data; and (5) multimodal methods combining multiple sensing technologies.

**Limitations of search strategy.** Several constraints should be noted. Our search was limited to English-language publications, potentially missing relevant work from regions with strong agricultural research programs publishing in other languages. The focus on peer-reviewed journal articles excluded potentially valuable findings in conference proceedings and technical reports. Additionally, the rapidly evolving nature of deep learning research means some very recent advances may not yet appear in the indexed literature.

## Deep Learning Architectures for Plant Disease Detection

Deep learning has transformed plant disease detection by allowing automatic extraction of features directly from raw images, removing the need for manual feature engineering (Ferentinos, 2018). For readers less familiar with AI terminology, a deep learning architecture refers to the specific design and arrangement of computational layers within an artificial neural network. Each architecture employs different strategies for processing input data such as convolutions for local feature extraction in Convolutional Neural Networks (CNNs) or self-attention mechanisms for capturing global relationships in Vision Transformers. The choice of architecture significantly influences model performance, computational requirements, and suitability for deployment in different agricultural settings. This section reviews the major deep learning architectures that have been applied to plant disease identification.

**CNNs.** CNNs are a class of deep learning models specifically designed for processing grid-structured data such as images. CNNs use learnable filters (also called kernels) that systematically slide across the input image to detect local patterns such as edges, textures, color gradients, and shapes at progressively higher levels of abstraction. A typical CNN consists of three types of layers: convolutional layers for hierarchical feature detection, pooling layers

for reducing spatial dimensions while retaining important information, and fully connected layers for producing the final disease classification. This hierarchical feature learning capability makes CNNs particularly effective for visual recognition tasks, including distinguishing between healthy tissue and various disease symptoms on plant leaves.

**AlexNet and VGGNet.** AlexNet (Krizhevsky et al., 2012) introduced key innovations such as ReLU activations, dropout regularization, and GPU-based training, achieving a top 5 error rate of 15.3% on ImageNet. In plant disease detection, AlexNet has been successfully applied through transfer learning, reaching 95–99% accuracy on the PlantVillage dataset (Ferentinos, 2018; Mohanty et al., 2016). VGGNet (Simonyan and Zisserman, 2014) showed that increasing network depth improves classification performance using uniform  $3 \times 3$  convolutions. For example, Sujatha et al. (2025) combined VGG19 with k-Nearest Neighbors to achieve 99.1% accuracy on the Custard Apple dataset. However, VGGNet's large size, with 138 million parameters, can limit deployment on mobile or edge devices.

**ResNet and DenseNet.** Residual Networks (ResNet) (He et al., 2016) use skip connections to overcome the degradation problem in very deep networks, enabling training of over 150 layers while achieving a 3.57% top 5 error on ImageNet. DenseNet (Huang et al., 2017) further improves feature reuse through dense connections between layers. Too et al. (2019) achieved 99.75% accuracy using DenseNet-121, showing consistent performance without overfitting, making it suitable for scenarios with limited training data.

**EfficientNet.** EfficientNet (Tan and Le, 2019) introduced a compound scaling method that balances network depth, width, and input resolution. Atila et al. (2021) compared EfficientNet variants (B0–B7) with other architectures including VGG16, VGG19, ResNet50, InceptionV3, MobileNetV2, and NASNetLarge on the PlantVillage dataset. EfficientNet-B5 achieved 99.91% accuracy on the original dataset and 99.97% on augmented data, with a precision of 99.39%, outperforming all other models. Its computational efficiency makes EfficientNet highly suitable for mobile and edge-device deployment in real-world agricultural applications.

**ViTs.** Vision Transformers represent a fundamentally different paradigm for image analysis compared to CNNs. Rather than using convolutional filters to detect local features, ViTs divide an input image into fixed-size patches (typically  $16 \times 16$  pixels), project each patch into a numerical vector (embedding), and process these embeddings

through transformer layers that use self-attention mechanisms. Self-attention allows the model to compute the relevance of every image patch relative to every other patch, thereby capturing long-range spatial dependencies and global contextual information that CNNs may miss. This global receptive field is particularly advantageous for plant disease detection, where disease symptoms may manifest across multiple regions of a leaf and the spatial relationships between symptoms can be diagnostically informative.

**Standard vision transformers.** ViTs (Dosovitskiy, 2020) use self-attention mechanisms applied to image patches, enabling the model to capture long-range dependencies across the entire image. Borhani et al. (2022) compared ViT architecture with traditional CNNs and hybrid models, showing that attention mechanisms improve accuracy but can slow inference. Using ViTs with optimized parameter counts, Singh et al. (2024) reported 99.92% accuracy on the PlantVillage dataset and 75.72% on PlantDoc, demonstrating strong performance across both controlled and real-world data.

**Swin transformer.** The Swin Transformer (Liu et al., 2021) introduces hierarchical architecture and shifted window self-attention to improve efficiency and adaptability across vision tasks. It achieves linear computational complexity while maintaining high performance, with 87.3% top 1 accuracy on ImageNet, 58.7 box AP on COCO, and 53.5 mIoU on ADE20K. Such performance makes it well-suited for plant disease detection. Liu and Zhang (2025) further developed efficient Swin Transformer variants tailored for deployment in agricultural IoT environments.

**Lightweight architectures.** Recent efforts have focused on creating lightweight transformer models for mobile and edge applications. Thai et al. (2023) developed Former Leaf for cassava disease detection, incorporating Least Important Attention Pruning (LeIAP) to reduce model size by 28% and accelerate evaluation by 15%, while achieving a 3% accuracy enhancement over the base model, demonstrating that pruning can simultaneously improve efficiency and maintain competitive performance. Li et al. (2023) proposed PMVT (Plant-based MobileViT), combining MobileViT with inverted residual blocks and CBAM attention modules for efficient smartphone deployment. PMVT achieved competitive accuracy with only 1.9 million parameters, making it one of the most parameter-efficient models for plant disease detection and particularly suitable for real-time inference on resource-constrained agricultural unmanned aerial vehicles (UAVs) and mobile devices. Lu et al. (2022) achieved 98.14% accuracy in grape disease detection using ghost modules within ViT encoders, demonstrating the potential for lightweight, high-performance

transformer-based models in real-world agricultural applications.

**Object detection architectures.** Object detection architectures are designed to locate diseased regions within images, which is particularly useful for field images containing multiple leaves, overlapping diseases, or complex backgrounds (Fuentes et al., 2017).

**Faster R-CNN.** Faster R-CNN (Ren et al., 2017) incorporates Region Proposal Networks that share convolutional features, enabling near real-time detection. Fuentes et al. (2017) compared Faster R-CNN, R-FCN, and SSD for tomato disease detection using VGGNet and ResNet as feature extractors, finding that two-stage detectors achieved higher mean Average Precision (mAP) for nine disease and pest types, even with inter- and intra-class variations. Arsenovic et al. (2019) further reported that Faster R-CNN with top-down modulation performed best when detecting multiple diseases in complex field conditions.

**YOLO family.** The YOLO (You Only Look Once) family emphasizes real-time detection through a single-pass architecture (Redmon et al., 2016). Arsenovic et al. (2019) applied YOLOv3 for leaf detection in the PlantDiseaseNet dataset. Wang et al. (2023) introduced YOLOv7, achieving 56.8% Average Precision at over 30 frames per second, making it suitable for UAV-based monitoring and rapid video processing. More recent YOLOv8 variants have been optimized for plant disease detection, addressing challenges such as small lesion sizes and class imbalance (Li et al., 2023).

**Performance metrics in plant disease detection.** Understanding the metrics used to evaluate deep learning models is essential for interpreting reported results and comparing studies. Accuracy represents the proportion of correctly classified images out of all test images; while intuitive, it can be misleading when class distributions are imbalanced, as a model that simply predicts the majority class can still achieve high accuracy. Precision measures the proportion of positive predictions that are correct (i.e., of all images the model classified as “diseased,” how many were truly diseased), while recall (also termed sensitivity) measures the proportion of actual positive cases that were correctly identified. The F1-score provides the harmonic mean of precision and recall, offering a balanced measure when both false positives (healthy plants misidentified as diseased) and false negatives (diseased plants missed) are important. mAP, commonly used in object detection tasks, summarizes precision-recall performance across multiple disease classes and intersection-over-union thresholds.

**Table 1.** Summary of deep learning and object detection architectures for plant disease detection

| Architecture      | Reference              | Dataset       | Classes | Accuracy (%)   | Precision (%) | Key contribution    | Category           |
|-------------------|------------------------|---------------|---------|----------------|---------------|---------------------|--------------------|
| EfficientNet-B5   | Atila et al., 2021     | PlantVillage  | 39      | 99.97          | 99.39         | Compound scaling    | CNN                |
| DenseNet-121      | Too et al., 2019       | PlantVillage  | 38      | 99.75          | -             | Dense connectivity  | CNN                |
| ViT               | Singh et al., 2024     | PlantVillage  | 38      | 99.92          | -             | Self-attention      | Vision Transformer |
| CNN               | Ferentinos, 2018       | 87,848 images | 58      | 99.53          | -             | Multi-species       | CNN                |
| VGG19 + kNN       | Sujatha et al., 2025   | Custard Apple | 4       | 99.10          | 99.10         | Hybrid ML + DL      | Hybrid             |
| ResNet-9 + SHAP   | Shafik et al., 2025    | TPPD          | 15      | 97.40          | 96.40         | Explainable AI      | CNN + XAI          |
| Siamese + Triplet | Argüeso et al., 2020   | PlantVillage  | 6       | 90.00          | -             | 89% data reduction  | Few-Shot           |
| PlantDiseaseNet   | Arsenovic et al., 2019 | 79,265 images | 42      | 93.67          | -             | Real-world dataset  | CNN                |
| Faster R-CNN      | Fuentes et al., 2017   | Tomato        | 9       | High           | -             | Object detection    | Detection          |
| Transfer Learning | Ramcharan et al., 2017 | Cassava field | 5       | 93–98          | -             | Field deployment    | Transfer           |
| FormerLeaf        | Thai et al., 2023      | Cassava       | 5       | 3% enhancement | -             | Attention pruning   | Lightweight ViT    |
| PMVT              | Li et al., 2023        | PlantVillage  | 38      | Competitive    | -             | 1.9M params; mobile | Lightweight ViT    |

ViTs, Vision Transformers; CNNs, Convolutional Neural Networks; PMVT, Plant-based Mobile ViT.

These metrics are reported throughout this review to enable meaningful comparison across studies. The summary of the deep learning and object detection architectures for plant disease detection is shown in Table 1.

Beyond accuracy, several complementary metrics are essential for rigorous evaluation of plant disease detection models, particularly given the class imbalance frequently observed in agricultural datasets where healthy samples vastly outnumber diseased ones. Sensitivity (also called recall or true positive rate) measures the proportion of truly diseased samples that the model correctly identifies; in a plant pathology context, low sensitivity means diseased plants are missed, potentially allowing epidemics to spread undetected. Specificity (true negative rate) measures the proportion of truly healthy plants correctly classified as healthy; low specificity results in false alarms that may trigger unnecessary and costly pesticide applications. The Receiver Operating Characteristic Area Under the Curve (ROC-AUC) provides a threshold-independent measure of a model's ability to discriminate between healthy and diseased classes across all possible decision boundaries, with values closer to 1.0 indicating superior discrimination and 0.5 indicating performance no better than random

chance. Confusion matrices offer a comprehensive tabular visualization of classification outcomes by displaying the frequency of correct and incorrect predictions for every disease class, enabling researchers to identify specific disease pairs that the model frequently confuses such as early blight and late blight on tomato, which share similar visual symptoms in early stages. Training and validation loss curves are also informative: a divergence between training loss (decreasing) and validation loss (increasing or plateauing) signals overfitting, where the model memorizes training images rather than learning generalizable disease features. For plant pathologists evaluating AI diagnostic tools, these metrics collectively provide a far more complete and clinically relevant assessment than accuracy alone. We recommend that future studies routinely report precision, recall, F1-score, specificity, and ROC-AUC, along with confusion matrices and loss curves, to enable rigorous comparison across studies and to ensure that models perform reliably across all disease classes, including rare but agronomically critical conditions.

Fig. 2 illustrates the evolution of deep learning architectures for plant disease detection from 2012 to 2025. Each column represents a distinct architectural generation, with

|                   | 2012                                                                              | 2015                                                                              | 2016                                                                              | 2019                                                                               | 2020–2023                                                                           | 2023–2025                                                                           |
|-------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Architecture Name | AlexNet                                                                           | VGGNet                                                                            | ResNet                                                                            | EfficientNet                                                                       | ViT/Swin                                                                            | Lightweight Models                                                                  |
| Model Icon        |  |  |  |  |  |  |
| Parameter Count   | 60M Params                                                                        | 138M Params                                                                       | 25M Params                                                                        | 30M Params                                                                         | 86M Params                                                                          | Optimized                                                                           |
| Key Innovation    | <i>ImageNet Breakthrough</i>                                                      | <i>Increased Depth</i>                                                            | <i>Skip Connections</i>                                                           | <i>Compound Scaling</i>                                                            | <i>Self-Attention</i>                                                               | <i>Edge Deployment</i>                                                              |

**Fig. 2.** Evolution of deep learning architectures for plant disease detection (2012–2025).

row headers indicating: Architecture Name, Model Icon, Parameter Count, and Key Innovation. Early models, such as AlexNet (2012), marked the breakthrough of deep learning in image classification, while deeper architectures like VGGNet (2015) and ResNet (2016) improved performance through increased depth and skip connections. EfficientNet (2019) introduced compound scaling for better accuracy and efficiency, followed by ViT and Swin Transformers (2020–2023), which leveraged self-attention mechanisms to capture global image features. Recent lightweight and optimized models (2023–2025) focus on edge deployment, enabling real-time inference on mobile and IoT devices. The number of parameters for each model reflects increasing complexity and computational demand over time.

### Learning Paradigms and Training Strategies

**Transfer learning.** Transfer learning has become a widely used strategy in plant disease detection, particularly when labeled agricultural data are limited. By reusing features learned from large-scale datasets such as ImageNet, models can be efficiently adapted to crop disease recognition tasks (Ferentinos, 2018; Mohanty et al., 2016). Low-level visual features, including edges, textures, and color patterns, transfer effectively across domains, while higher-level layers are fine-tuned to capture disease-specific characteristics.

Ferentinos (2018) demonstrated the effectiveness of this

approach by training CNN models on 87,848 images spanning 25 plant species and 58 plant–disease combinations, achieving an overall accuracy of 99.53%. Similarly, Ramcharan et al. (2017) applied transfer learning to cassava disease detection using real field images collected in Tanzania and reported accuracies ranging from 93% to 98% across five disease classes. However, Barbedo (2018) showed that performance can vary substantially depending on crop type, imaging conditions, and disease symptoms, with reported accuracies ranging from 60% to 100%.

**Few-shot learning.** Few-shot learning aims to recognize new disease classes using only a small number of labeled examples, which is particularly valuable for rare or emerging plant diseases (Argüeso et al., 2020). Unlike traditional transfer learning approaches that typically require hundreds of images per class, few-shot learning methods can operate effectively with as few as one to twenty samples.

Argüeso et al. (2020) proposed a metric-based few-shot learning framework using Siamese networks with triplet loss. Experiments on the PlantVillage dataset, divided into source and unseen target classes, achieved median accuracies of 55.5% with one image per class, 80.0% with 15 images, and 90.0% with 80 images. This corresponds to an 89.1% reduction in training data while incurring only a small loss in accuracy. In contrast, conventional fine-tuning

approaches achieved substantially lower performance under the same low-data conditions, highlighting the advantage of metric learning for data-scarce scenarios.

**Self-supervised learning.** Self-supervised learning (SSL) addresses the challenge of limited labeled data by learning meaningful representations directly from unlabeled images (Huan et al., 2025). Huan et al. proposed a unified SSL framework that combines BYOL, masked image modeling, and contrastive learning within a ResNet101 backbone. The model achieved 77.82% accuracy on the PlantDoc dataset, along with 80.00% precision and 78.24% recall. When fine-tuned on PlantVillage, performance increased to 99.85% accuracy, demonstrating strong generalization across datasets and highlighting SSL as a scalable and annotation-efficient solution for agricultural applications.

**Data augmentation strategies.** Data augmentation is commonly used to improve model robustness and address class imbalance in plant disease datasets. Arsenovic et al. (2019) compared conventional augmentation techniques, such as rotation, flipping, and scaling, with generative adversarial network (GAN)-based image synthesis. Their results

showed that StyleGAN-generated images preserved critical disease features, including lesion shape, color, and texture, leading to improved classification performance. In a related study, Shafik et al. (2025) employed extensive augmentation strategies to balance class distributions in the TPPD dataset, contributing to an overall accuracy of 97.4% across 15 disease classes.

Table 2 summarizes key learning approaches for plant disease detection. Transfer learning offers the highest accuracy but requires large, labeled datasets, while few-shot learning reduces data needs with moderate accuracy. Self-supervised learning operates without labeled data, making it suitable for large-scale applications with limited annotations.

**Benchmark Datasets for Plant Disease Detection**

The availability of large-scale, well-annotated datasets has been fundamental to progress in AI-based plant disease detection. Table 3 summarizes the major publicly available datasets, their characteristics, and limitations.

**Background bias in the PlantVillage dataset.** While the

**Table 2.** Comparison of learning paradigms for plant disease detection

| Approach          | Images needed       | Accuracy | Best used when                                         |
|-------------------|---------------------|----------|--------------------------------------------------------|
| Transfer learning | 500–1,000 per class | 99%+     | Large labeled datasets available                       |
| Few-shot learning | 5–80 per class      | 80–90%   | Very limited images available                          |
| Self-supervised   | Many unlabeled      | 77–99%   | No expert labels available; large-scale unlabeled data |

**Table 3.** Publicly available datasets for plant disease detection research

| Dataset         | Images  | Classes/Crops | Imaging Environment                  | Annotation Type | Key Limitation                            | Reference                |
|-----------------|---------|---------------|--------------------------------------|-----------------|-------------------------------------------|--------------------------|
| PlantVillage    | 54,303  | 38/14 crops   | Controlled lab (uniform backgrounds) | Class labels    | Background bias; not field-representative | Hughes and Salathé, 2015 |
| PlantDoc        | 2,598   | 27/13 plants  | Real-world field conditions          | Class + BBox    | Small size; limited disease diversity     | Singh et al., 2020       |
| PlantDiseaseNet | 79,265  | 42/12 species | Natural field conditions             | Class + BBox    | Limited geographic diversity              | Arsenovic et al., 2019   |
| TPPD (Turkey)   | 4,447   | 15/6 plants   | Mixed (lab + field images)           | Class labels    | Regional focus; moderate size             | Shafik et al., 2025      |
| Cassava         | ~21,000 | 5/1 crop      | Field (Tanzania)                     | Class labels    | Single crop; regional specificity         | Ramcharan et al., 2017   |
| Tomato          | ~18,000 | 9/1 crop      | Controlled greenhouse                | BBox + Class    | Single crop; greenhouse only              | Fuentes et al., 2017     |

BBox, Bounding box annotations.  
“Imaging Environment” describes the physical conditions under which images were captured. “Controlled lab” indicates uniform artificial backgrounds; “Field” indicates natural agricultural settings with complex backgrounds; “Mixed” indicates a combination of both conditions. “Key Limitation” summarizes the primary constraint of each dataset for generalizability.

PlantVillage dataset has been instrumental in advancing deep learning research for plant disease detection, it is critical to acknowledge the inherent background bias in this dataset. PlantVillage images were captured under controlled laboratory conditions with uniform, plain backgrounds that do not represent the visual complexity of real agricultural environments. Noyan (2022) conducted a rigorous analysis demonstrating that deep learning models trained on PlantVillage frequently learn to associate background features rather than actual disease symptoms with particular disease classes. When tested on images with altered or removed backgrounds, model accuracy decreased dramatically, revealing that reported high-accuracy results may substantially overestimate true diagnostic capability.

This finding has several significant implications. First, models reporting near-perfect accuracy on PlantVillage should not be assumed to generalize to field conditions without explicit validation on independent, field-collected datasets such as PlantDoc (Singh et al., 2020) or PlantDiseaseNet (Arsenovic et al., 2019). Second, it highlights the pressing need for training datasets that include diverse, naturalistic backgrounds representative of actual farming environments. Third, it underscores the importance of using explainability techniques such as Grad-CAM (discussed in Section 7) to verify that models are attending to disease-relevant features rather than background artifacts. Researchers and practitioners should exercise caution when interpreting PlantVillage-based results and should prioritize

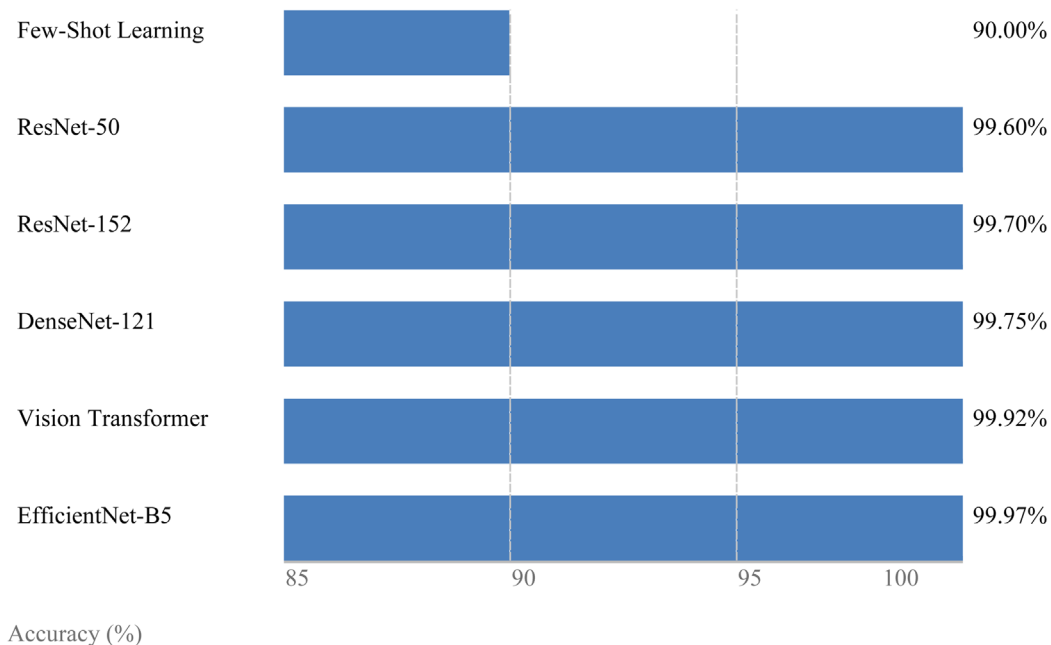
evaluation on field-representative datasets for any claims of practical applicability.

Fig. 3 compares the classification accuracy of different deep learning models on the PlantVillage dataset. Modern CNNs and vision transformer models achieve very high accuracy above 99%, with EfficientNet-B5 showing the best performance. Few-shot learning attains lower accuracy but requires substantially fewer training data, highlighting the trade-off between performance and data efficiency.

### Imaging Modalities and Sensor Technologies

**Red, Green, Blue (RGB) Imaging.** RGB imaging is the most widely used modality for plant disease detection because of its low cost and easy access through smartphones (Hughes and Salathé, 2015). Early studies by Camargo and Smith (2009) used color transformations and histogram-based segmentation to identify disease symptoms from RGB images. Due to its simplicity, RGB imaging remains the foundation of many mobile diagnostic tools, especially for smallholder farmers in regions lacking advanced equipment.

**Hyperspectral imaging.** Hyperspectral imaging captures detailed spectral information across many wavelength bands, allowing detection of plant stress before visible symptoms appear (Lowe et al., 2017; Mahlein et al., 2018). Rumpf et al. (2010) successfully identified sugar beet diseases using hyperspectral reflectance and support vector



**Fig. 3.** Comparison of deep learning model accuracy on PlantVillage dataset.

machines, achieving up to 97% accuracy for visible symptoms and 65–90% for early-stage detection. Zhang et al. (2019) further showed that combining UAV-based hyperspectral data with deep CNNs improved yellow rust detection in wheat, outperforming traditional machine learning approaches.

**Multimodal sensor fusion.** Combining multiple sensing technologies can improve early disease detection by capturing different plant responses. Berdugo et al. (2014) integrated hyperspectral imaging, chlorophyll fluorescence, and infrared thermography to detect cucumber diseases. Each sensor contributed complementary physiological information, and their fusion enabled reliable presymptomatic disease identification.

**UAV-based remote sensing.** UAVs provide efficient, large-scale crop monitoring under real field conditions (Bouguettaya et al., 2023). Kerkech et al. (2020) used UAV-based visible and infrared imagery with deep learning segmentation to detect grapevine mildew. Their approach achieved high detection accuracy at both leaf and plant levels, supporting targeted disease management in precision agriculture.

## Explainable AI and Model Interpretability

As deep learning models are increasingly used in agricultural decision-making, explainable artificial intelligence (XAI) has become important for improving trust, validating model behavior, identifying errors, and gaining biological insight into disease symptoms (Selvaraju et al., 2017; Brahimi et al., 2018; Shafik et al., 2025).

**Gradient-based visualization.** Gradient-weighted Class Activation Mapping (Grad-CAM) highlights image regions that most influence model predictions by tracing gradients from the final convolutional layers (Selvaraju et al., 2017). In plant disease detection, Grad-CAM helps confirm that models focus on meaningful disease features such as lesions or discoloration rather than background noise. Brahimi et al. (2018) used saliency maps to interpret CNN predictions while achieving high classification accuracy, showing that deep models can also support visual understanding of disease symptoms. From a plant pathological perspective, Grad-CAM visualizations offer a direct means of verifying whether AI models identify diagnostically meaningful features. For example, Grad-CAM can be used to confirm whether a model correctly identifies the chlorotic halo characteristic of certain bacterial infections, such

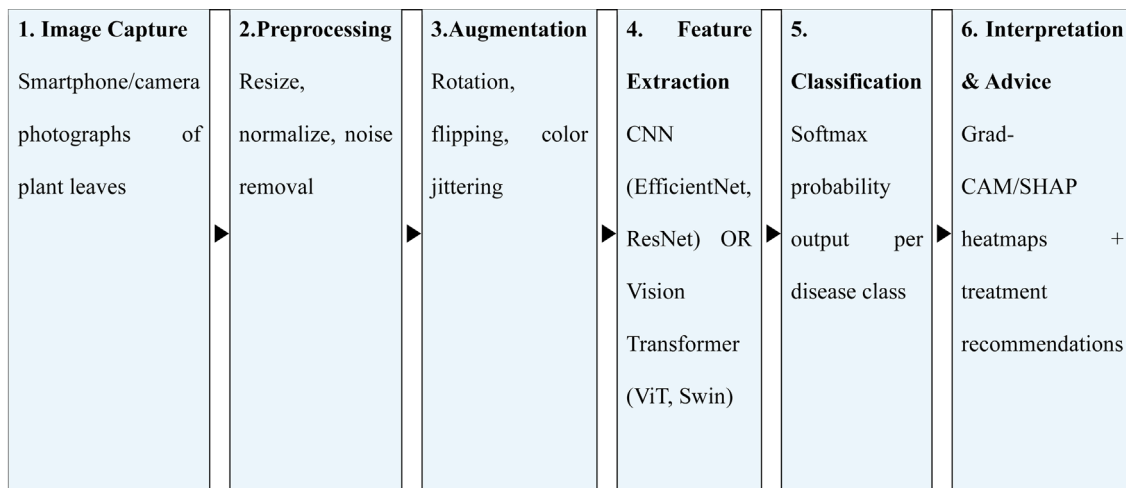
as bacterial leaf blight caused by *Xanthomonas oryzae*, or whether it focuses on the concentric ring patterns typical of *Alternaria* leaf spot. Brahimi et al. (2018) demonstrated that their CNN model consistently highlighted lesion boundaries and symptomatic tissue in tomato leaves, providing visual confirmation that the learned features align with established Phyto pathological knowledge. Similarly, Shafik et al. (2025) used Grad-CAM alongside SHAP to show that their ResNet-9 model attended to texture changes and lesion margins rather than background regions or healthy tissue. Such pathologically grounded validation is essential for building trust among plant pathologists and ensuring that AI models can serve as reliable diagnostic support tools rather than opaque black boxes.

**SHAP-based interpretability.** SHapley Additive exPlanations (SHAP) provide a quantitative way to interpret model decisions by measuring the contribution of different features (Shafik et al., 2025). Using SHAP with a ResNet-9 model, Shafik et al. achieved 97.4% accuracy on the TPPD dataset and showed that predictions relied on lesion boundaries, texture and color changes, and other disease-related cues. Such interpretable outputs allow experts to validate model decisions and identify potential biases before field deployment.

**Bridging XAI and plant pathology practice.** The integration of explainability techniques with domain-specific pathological knowledge represents a critical frontier for making AI-driven plant disease detection more relevant and trustworthy for plant pathologists. Beyond simply highlighting image regions, XAI techniques can be used to validate whether models have learned biologically meaningful disease signatures. For instance, a model that correctly identifies bacterial blight should show high activation over water-soaked lesion margins rather than leaf veins or petioles. Future work should establish standardized protocols for XAI-based model validation in plant pathology, where explainability outputs are systematically compared against known symptom descriptions from disease compendia and expert annotations. Such protocols would facilitate the adoption of AI tools in diagnostic laboratories and extension services, where model transparency is a prerequisite for practical use.

## Overview of the Proposed Deep Learning Framework for Plant Disease Detection and Diagnosis

Fig. 4 illustrates the overall workflow of a deep learning-based plant disease detection and diagnosis system. The



**Fig. 4.** Schematic workflow of the proposed deep learning-based plant disease detection and diagnosis system. Note: CNN architectures (e.g., EfficientNet, ResNet) and Vision Transformer architectures (e.g., ViT, Swin Transformer) are correctly represented as distinct model families in the feature extraction stage.

workflow begins with image capture of plant leaves, followed by preprocessing and data augmentation to improve image quality and dataset diversity. Deep learning models including both CNN architectures (e.g., EfficientNet, ResNet) and Vision Transformer architectures (e.g., ViT, Swin Transformer), which are distinct model families are then used for feature extraction and disease classification, producing confidence-based predictions. Finally, model interpretation techniques provide visual explanations, and the system delivers actionable treatment and management recommendations based on the detected disease.

### Current Challenges and Limitations

**Domain gap between laboratory and field.** A major limitation of AI-based plant disease detection is the reduced performance of models when transferred from controlled laboratory settings to real field conditions. Mohanty et al. (2016) first documented a decline in accuracy from nearly 99% to 31% on external images. This domain gap is primarily caused by complex field backgrounds, variable lighting, inconsistent image quality from mobile devices, and the diverse expression of disease symptoms under combined biotic and abiotic stresses (Arsenovic et al., 2019). Noyan (2022) further demonstrated that background bias in commonly used laboratory-captured datasets exacerbates this problem, as models may learn to classify based on imaging conditions rather than genuine disease features. Domain adaptation techniques, including adversarial training (Ganin et al., 2016) and style transfer methods, represent promising but still underexplored approaches to

mitigating this gap in the plant pathology context.

**Data scarcity and class imbalance.** Although datasets such as PlantVillage contain over 54,000 images, they represent only a small subset of the more than 30,000 cultivated plant species and their diseases worldwide (Barbedo, 2018). Many economically important diseases, particularly in tropical regions, remain underrepresented. In addition, class imbalance—where healthy samples dominate—biases models toward majority classes and reduces detection of rare but critical diseases. Strategies such as synthetic data generation using GANs (Arsenovic et al., 2019), cost-sensitive learning, and oversampling techniques (Shafik et al., 2025) have shown promise in addressing these imbalances but require further validation across diverse crop systems.

### Computational constraints and edge deployment.

Modern deep learning models often require substantial computational resources, limiting deployment on edge devices such as smartphones and agricultural drones. While efficient architectures like EfficientNet have improved feasibility, real-time performance on resource-constrained platforms still depends on further optimization through pruning, quantization, and model compression. The development of lightweight models such as PMVT (Li et al., 2023), with only 1.9 million parameters, and pruned transformers such as FormerLeaf (Thai et al., 2023) demonstrates that competitive accuracy can be achieved with significantly reduced computational overhead, enabling deployment on agricultural UAVs and embedded devices

for real-time field monitoring.

**Multi-disease detection.** In real-world agriculture, plants frequently exhibit symptoms of multiple diseases simultaneously. However, most existing models assume single-disease classification. Object detection approaches have shown promise in addressing this limitation (Fuentes et al., 2017), but accurate and scalable multi-disease detection remains a challenge. Future models should incorporate multi-label classification and hierarchical disease taxonomies to more accurately reflect the co-occurrence of diseases under field conditions.

**Standardization and reproducibility.** Furthermore, given the prevalence of class imbalance in plant disease datasets, we strongly recommend that future studies report not only overall accuracy but also precision, recall, F1-score, sensitivity, specificity, and ROC-AUC values for each disease class. Confusion matrices should be provided to reveal inter-class misclassification patterns, and training/validation loss curves should be included to enable assessment of model convergence and potential overfitting. Adopting these comprehensive evaluation practices will strengthen the rigor, reproducibility, and clinical relevance of AI-based plant disease detection research and facilitate more meaningful cross-study comparisons.

The absence of standardized evaluation protocols hinders fair comparison across studies. Differences in data splits, training strategies, and performance metrics reduce reproducibility. Establishing common benchmarks and reporting standards is essential for reliable progress in this field. We recommend that future studies adopt standardized dataset splits, report multiple metrics (accuracy, precision, recall, F1-score), and publicly release code and model weights to facilitate reproducibility and independent verification.

## Future Research Directions

**Foundation models and large-scale pre-training.** Recent advances in vision-language foundation models, such as CLIP (Radford et al., 2021), and large-scale segmentation models such as the Segment Anything Model (Kirillov et al., 2023), have demonstrated remarkable zero-shot and few-shot generalization in general computer vision tasks, suggesting strong potential for adaptation to agricultural applications. Bommasani et al. (2021) formally characterized foundation models and highlighted their capacity for cross-domain transfer through large-scale pre-training.

The success of foundation models in natural language processing highlights their potential in agricultural AI. Large-scale pre-training on diverse plant imagery, includ-

ing unlabeled data through self-supervised learning, could generate robust visual representations that adapt efficiently to specific diseases with limited labeled data. Vision-language models that integrate plant images with textual disease descriptions represent a promising research direction. Such models could enable zero-shot disease classification, where a model identifies diseases it has never seen during training by leveraging textual symptom descriptions, potentially transforming diagnostic capability for rare and emerging pathogens.

**Federated learning for privacy-preserving collaboration.** McMahan et al. (2017) introduced the foundational Federated Averaging (FedAvg) algorithm, which enables decentralized model training by aggregating locally computed gradient updates without centralizing raw data. Durrant et al. (2022) reviewed the application of federated learning in smart agriculture and demonstrated its feasibility for training crop disease models across geographically distributed farms while maintaining data sovereignty. Nguyen et al. (2021) further showed that federated approaches can achieve classification performance comparable to centralized training while preserving privacy in sensitive IoT-based agricultural monitoring systems.

Agricultural datasets often contain sensitive or proprietary information related to crop management. Federated learning enables collaborative model training across distributed data sources without sharing raw data, allowing cooperation among farms, research institutions, and industry while preserving data privacy and intellectual property. This approach is particularly relevant for building globally representative disease detection models, as individual farms or regions may have limited disease diversity, and federated aggregation can combine locally learned features into more robust global models.

**Integration with IoT and precision agriculture.** Wolfert et al. (2017) provided an early comprehensive analysis of big data applications in smart farming, identifying real-time sensor integration and automated decision support as key requirements for precision agriculture systems. Kamilaris and Prenafeta-Boldú (2018) reviewed deep learning applications in agriculture broadly, confirming that integration of image-based AI with environmental sensor networks represents a critical pathway toward predictive crop management.

Future plant disease detection systems are expected to integrate with precision agriculture frameworks by combining image-based diagnosis with data from environmental sensors, soil measurements, weather forecasts, and crop

growth models. Such multimodal systems can support predictive disease management by identifying risk factors before visible symptoms appear. A critical enabler for IoT-based disease detection is the availability of lightweight models that can operate within the computational and energy constraints of edge devices. Models such as PMVT (Li et al., 2023), with only 1.9 million parameters, demonstrate that competitive diagnostic accuracy can be maintained at a fraction of the computational cost of standard architectures such as VGGNet (138M parameters) or standard ViT (86M parameters). Such low-parameter models are essential for enabling real-time inference on agricultural UAVs, which must balance payload weight, battery life, and processing speed during field surveillance missions. Similarly, pruned transformer variants such as FormerLeaf (Thai et al., 2023), which reduced model size by 28% while achieving a 3% accuracy enhancement, and ghost-module-based architectures (Lu et al., 2022) offer viable pathways to deploy high-performance models on embedded sensors integrated into irrigation systems, automated greenhouses, and robotic platforms.

The value and importance of lightweight models extends beyond mere computational convenience. In resource-limited agricultural contexts including smallholder farms in developing regions where internet connectivity is unreliable and cloud computing is impractical, on-device inference is the only viable deployment strategy. Lightweight models enable smartphone-based diagnostics that can function entirely offline, providing immediate disease identification and treatment recommendations to farmers who lack access to expert pathologists or laboratory facilities. Furthermore, the integration of lightweight detection models with IoT sensor networks (temperature, humidity, soil moisture) can enable predictive disease management systems that trigger alerts before visible symptoms emerge. The development and systematic validation of lightweight models specifically optimized for agricultural edge deployments should therefore be a high priority for future research, as the practical adoption of AI-driven precision agriculture depends critically on the ability to run accurate models efficiently at the point of data collection.

**Robotic systems and autonomous management.** Duckett et al. (2018) provided a comprehensive roadmap for agricultural robotics, identifying real-time disease detection as a key enabling capability for autonomous crop management systems. Bac et al. (2014) demonstrated that integrating computer vision with robotic harvesting platforms can achieve precision treatment at the individual plant level, reducing chemical inputs while maintaining crop health.

Fountas et al. (2020) reviewed the current state of agricultural robotics for field operations and emphasized that on-board AI inference capabilities are essential for effective autonomous monitoring under variable field conditions.

The integration of AI-based disease detection with agricultural robotics offers opportunities for autonomous monitoring and targeted treatment. Real-time detection combined with robotic actuation could enable localized intervention, reducing pesticide use and supporting more sustainable crop management practices. For robotic platforms operating in real-time field conditions, model inference latency and computational efficiency are as important as classification accuracy. Lightweight architectures specifically designed for edge deployment, such as PMVT and pruned transformers, are particularly relevant for integration into autonomous agricultural robots and UAV-based surveillance systems that require on-board inference without cloud connectivity.

**Citizen science and crowdsourced data.** Platforms such as Pl@ntNet (Joly et al., 2016) and iNaturalist (Van Horn et al., 2018) have demonstrated the feasibility of large-scale crowdsourced botanical image collection, with Pl@ntNet accumulating millions of geotagged plant observations worldwide. Ceccaroni et al. (2019) reviewed frameworks for ensuring data quality in citizen science projects in the age of artificial intelligence, highlighting that expert validation combined with consensus-based annotation protocols can yield datasets of sufficient quality for training deep learning models. These findings suggest that community-contributed agricultural datasets, when properly curated, can serve as viable training resources for expanding disease detection capabilities to underrepresented crops and regions.

Mobile-based platforms that allow farmers to contribute geotagged disease images can significantly expand training datasets, particularly for underrepresented crops and regions. When combined with expert validation and active learning, crowdsourced data can accelerate model improvement while fostering engagement within farming communities.

**Towards a unified taxonomic framework for AI-based plant disease detection.** This need for systematic categorization aligns with calls by Boulent et al. (2019) for standardized benchmarking protocols in convolutional neural network-based crop disease identification, and builds on the hierarchical classification frameworks proposed in the broader computer vision literature (Khan et al., 2022). Establishing such a multi-dimensional taxonomy would also

facilitate systematic meta-analyses of model performance across diverse agricultural contexts and accelerate the identification of optimal architecture-paradigm combinations for specific deployment scenarios.

A significant gap in the current literature is the absence of a unified taxonomic framework that systematically classifies AI-based plant disease detection approaches along multiple dimensions, including architecture type (CNN, transformer, hybrid), learning paradigm (supervised, few-shot, self-supervised), deployment target (cloud, edge, mobile), and interpretability level (black-box, attention-visualizable, SHAP-explainable). Such a framework would enable more systematic comparison of approaches, identification of underexplored combinations, and strategic prioritization of research efforts. We propose that future work develop and validate such a multi-dimensional classification scheme, which could serve as a roadmap for both researchers entering the field and practitioners selecting appropriate models for specific agricultural contexts.

### Limitations of This Review

While this review provides a comprehensive synthesis of AI-based plant disease detection research, several limitations should be acknowledged:

**Language restriction:** Only English-language publications were included in this review, potentially excluding relevant studies published in other languages, particularly from non-English speaking regions with significant agricultural research output.

**Publication bias:** The review focused exclusively on peer-reviewed journal articles, excluding conference proceedings, preprints, and technical reports. This may introduce publication bias, as studies with negative or inconclusive results are less likely to be published in peer-reviewed venues.

**Temporal scope:** The review covers studies published between 2008 and 2025. Given the rapid advancement of deep learning technologies, some very recent developments may not be captured, and earlier foundational works using traditional machine learning approaches may be underrepresented.

**Single reviewer screening:** Due to resource constraints, the initial screening of titles and abstracts was conducted by a single reviewer, which may introduce selection bias. Dual independent screening with consensus resolution would have strengthened the review methodology.

**Limited crop diversity:** The majority of studies focused on a narrow range of crops (primarily tomato, potato, apple, and cassava), with limited representation of crops

important to food security in developing regions.

**Reproducibility concerns:** Many studies did not provide sufficient details on hyperparameter settings, data augmentation protocols, or source code availability. This lack of transparency limits the reproducibility of reported results and makes independent verification challenging.

### Conclusions

Artificial intelligence has substantially transformed plant disease detection, with deep learning and computer vision enabling accurate identification of plant diseases under controlled conditions. This review of 41 studies published between 2008 and 2025 highlights significant methodological progress, including the development of efficient convolutional network architectures, transformer-based models, few-shot learning frameworks, and multimodal sensing techniques capable of detecting disease before visible symptoms appear.

Despite these advancements, several obstacles remain for real-world application. Model performance often drops when moving from laboratory datasets to field conditions, a problem compounded by background bias in widely used benchmark datasets (Noyan, 2022). Representative datasets for rare or region-specific diseases are limited, computational demands challenge deployment on mobile and embedded devices, and standardization of evaluation protocols is still lacking. Addressing these issues is critical for the adoption of reliable, field-ready disease detection systems.

Future efforts should focus on enhancing model generalization in diverse field environments, creating lightweight and interpretable AI systems with particular emphasis on low-parameter models suitable for UAV and IoT deployment and integrating disease detection with precision agriculture technologies, including IoT sensors and autonomous platforms. Approaches such as foundation models, federated learning, and community-contributed datasets offer promising avenues to improve data availability, privacy, and scalability.

Given the growing pressures of climate change and global population growth, AI-based plant disease detection represents a vital tool for sustainable agriculture. Collaborative efforts across AI researchers, agronomists, engineers, and farming communities will be essential to translate research advances into effective, practical solutions for crop protection.

### Conflicts of Interest

No potential conflict of interest relevant to this article was

reported.

## References

- Argüeso, D., Picon, A., Irusta, U., Medela, A., San-Emeterio, M. G., Bereciartua, A. and Alvarez-Gila, A. 2020. Few-shot learning approach for plant disease classification using images taken in the field. *Comput. Electron. Agric.* 175:105542.
- Arsenovic, M., Karanovic, M., Sladojevic, S., Anderla, A. and Stefanovic, D. 2019. Solving current limitations of deep learning based approaches for plant disease detection. *Symmetry (Basel)* 11:939.
- Atila, Ü., Uçar, M., Akyol, K. and Uçar, E. 2021. Plant leaf disease classification using EfficientNet deep learning model. *Ecol. Inform.* 61:101182.
- Bac, C. W., van Henten, E. J., Hemming, J. and Edan, Y. 2014. Harvesting robots for high-value crops: state-of-the-art review and challenges ahead. *J. Field Robot.* 31:888-911.
- Barbedo, J. G. A. 2018. Impact of dataset size and variety on the effectiveness of deep learning and transfer learning for plant disease classification. *Comput. Electron. Agric.* 153:46-53.
- Barman, U., Sarma, P., Rahman, M., Deka, V., Lahkar, S., Sharma, V. and Saikia, M. J. 2024. ViT-SmartAgri: Vision transformer and smartphone-based plant disease detection for smart agriculture. *Agronomy* 14:327.
- Berdugo, C. A., Zito, R., Paulus, S. and Mahlein, A.-K. 2014. Fusion of sensor data for the detection and differentiation of plant diseases in cucumber. *Plant Pathology* 63:1344-1356.
- Bock, C. H., Poole, G. H., Parker, P. E. and Gottwald, T. R. 2010. Plant disease severity estimated visually, by digital photography and image analysis, and by hyperspectral imaging. *CRC Crit. Rev. Plant Sci.* 29:59-107.
- Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S. ... Liang, P. 2021. On the opportunities and risks of foundation models. *arXiv Preprint arXiv:2108.07258*.
- Borhani, Y., Khoramdel, J. and Najafi, E. 2022. A deep learning based approach for automated plant disease classification using vision transformer. *Sci. Rep.* 12:11554.
- Bouguettaya, A., Zarzour, H., Kechida, A. and Taberkit, A. M. 2023. A survey on deep learning-based identification of plant and crop diseases from UAV-based aerial images. *Cluster Comput.* 26:1297-1317.
- Boulent, J., Foucher, S., Théau, J. and St-Charles, P. L. 2019. Convolutional neural networks for the automatic identification of plant diseases. *Front. Plant Sci.* 10:941.
- Brahimi, M., Arsenovic, M., Laraba, S., Sladojevic, S., Boukhalfa, K. and Moussaoui, A. 2018. Deep learning for plant diseases: detection and saliency map visualisation. In: Human and machine learning, eds. J. Zhou and F. Chen, pp. 93-117. Springer, Cham, Switzerland.
- Camargo, A. and Smith, J. S. 2009. An image-processing based algorithm to automatically identify plant disease visual symptoms. *Biosyst. Eng.* 102:9-21.
- Ceccaroni, L., Bibby, J., Roger, E., Flemons, P., Michael, K., Fagan, L. and Oliver, J. L. 2019. Opportunities and risks for citizen science in the age of artificial intelligence. *Citiz. Sci. Theory Pract.* 4:29.
- Dosovitskiy, A. 2020. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv Preprint arXiv:2010.11929*.
- Duckett, T., Pearson, S., Blackmore, S., Grieve, B., Chen, W. H., Cielniak, G., ... Yang, G. Z. 2018. Agricultural robotics: the future of robotic agriculture. *arXiv Preprint arXiv:1806.06762*.
- Durrant, A., Markovic, M., Matthews, D., May, D., Enright, J. and Leontidis, G. 2022. The role of cross-silo federated learning in facilitating data sharing in the agri-food sector. *Comput. Electron. Agric.* 193:106648.
- Ferentinos, K. P. 2018. Deep learning models for plant disease detection and diagnosis. *Comput. Electron. Agric.* 145:311-318.
- Food and Agriculture Organization (FAO). 2023. Plant production and protection division. FAO, Rome, Italy.
- Fountas, S., Mylonas, N., Malounas, I., Rodias, E., Hellmann Santos, C. and Pekkeriet, E. 2020. Agricultural robotics for field operations. *Sensors (Basel)* 20:2672.
- Fuentes, A., Yoon, S., Kim, S. C. and Park, D. S. 2017. A robust deep-learning-based detector for real-time tomato plant diseases and pests recognition. *Sensors (Basel)* 17:2022.
- Ganin, Y., Ustinova, E., Ajakan, H., Germain, P., Larochelle, H., Laviolette, F., Marchand, M. and Lempitsky, V. 2016. Domain-adversarial training of neural networks. *J. Mach. Learn. Res.* 17:1-35.
- He, K., Zhang, X., Ren, S. and Sun, J. 2016. Deep residual learning for image recognition. In: Proc. IEEE Conf. Comput. Vis. Pattern Recognit., pp. 770-778. IEEE, New York, USA.
- Huan, X., Chen, B. and Zhou, H. 2025. A unified self-supervised framework for plant disease detection on laboratory and in-field images. *Electronics* 14:3410.
- Huang, G., Liu, Z., Van der Maaten, L. and Weinberger, K. Q. 2017. Densely connected convolutional networks. In: Proc. IEEE Conf. Comput. Vis. Pattern Recognit., pp. 4700-4708. IEEE, Honolulu, HI, USA.
- Hughes, D. P. and Salathé, M. 2015. An open access repository of images on plant health to enable the development of mobile disease diagnostics. *arXiv Preprint arXiv:1511.08060*.
- Joly, A., Bonnet, P., Goëau, H., Barbe, J., Selmi, S., Champ, J., Dufour-Kowalski, S., Affouard, A., Carré, J., Molino, J., Boujemaa, N. and Barthélémy, D. 2016. A look inside the Pl@ntNet experience. *Multimed. Syst.* 22:751-766.
- Kamilaris, A. and Prenafeta-Boldú, F. X. 2018. Deep learning in agriculture: A survey. *Comput. Electron. Agric.* 147:70-90.
- Kerkech, M., Hafiane, A. and Canals, R. 2020. Vine disease detection in UAV multispectral images using optimized image registration and deep learning segmentation approach. *Comput. Electron. Agric.* 174:105446.
- Khan, S., Naseer, M., Hayat, M., Zamir, S. W., Khan, F. S. and Shah, M. 2022. Transformers in vision: a survey. *ACM Com-*

- put. Surv.* 54:1-41.
- Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., et al. 2023. Segment anything. In: Proc. IEEE/CVF Int. Conf. Comput. Vis., pp. 4015-4026. IEEE, Paris, France.
- Krizhevsky, A., Sutskever, I. and Hinton, G. E. 2012. ImageNet classification with deep convolutional neural networks. In: Proc. Adv. Neural Inf. Process. Syst., pp. 1097-1105. Curran Associates, Inc., Red Hook, NY, USA.
- LeCun, Y., Bengio, Y. and Hinton, G. 2015. Deep learning. *Nature* 521:436-444.
- Li, G., Wang, Y., Zhao, Q., Yuan, P. and Chang, B. 2023. PMVT: a lightweight vision transformer for plant disease identification on mobile devices. *Front. Plant Sci.* 14:1256773.
- Liu, W. and Zhang, A. 2025. Plant disease detection algorithm based on efficient swin transformer. *Comput. Mater. Contin.* 82:3045-3068.
- Liu, Z., Lin, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S. and Guo, B. 2021. Swin transformer: Hierarchical vision transformer using shifted windows. In: Proc. IEEE/CVF Int. Conf. Comput. Vis., pp. 10012-10022. IEEE, Montreal, Canada.
- Lowe, A., Harrison, N. and French, A. P. 2017. Hyperspectral image analysis techniques for the detection and classification of the early onset of plant disease and stress. *Plant Methods* 13:80.
- Lu, X., Yang, R., Zhou, J., Jiao, J., Liu, F., Liu, Y., ... Gu, P. 2022. A hybrid model of ghost-convolution enlightened transformer for effective diagnosis of grape leaf disease and pest. *J. King Saud Univ. Comput. Inf. Sci.* 34:1755-1767.
- Mahlein, A. K. 2016. Plant disease detection by imaging sensors - parallels and specific demands for precision agriculture and plant phenotyping. *Plant Dis.* 100:241-251.
- Mahlein, A. K., Kuska, M. T., Behmann, J., Polder, G. and Walter, A. 2018. Hyperspectral sensors and imaging technologies in phytopathology: state of the art. *Annu. Rev. Phytopathol.* 56:535-558.
- McMahan, B., Moore, E., Ramage, D., Hampson, S. and y Arcas, B. A. 2017. Communication-efficient learning of deep networks from decentralized data. In: Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS 2017), eds. by A. Singh and J. Zhu, pp. 1273-1282. PMLR, Fort Lauderdale, FL, USA
- Mohanty, S. P., Hughes, D. P. and Salathé, M. 2016. Using Deep Learning for Image-Based Plant Disease Detection. *Front. Plant Sci.* 7:1419.
- Moher, D., Liberati, A., Tetzlaff, J. and Altman, D. G. 2009. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA Statement. *Open Med.* 3:e123-e130.
- Nguyen, D. C., Ding, M., Pathirana, P. N., Seneviratne, A., Li, J. and Poor, H. V. 2021. Federated learning for internet of things: A comprehensive survey. *IEEE Commun. Surv. Tutor.* 23:1622-1658.
- Noyan, M. A. 2022. Uncovering bias in the PlantVillage dataset. *arXiv preprint arXiv:2206.04374*.
- Nutter, F. W., Teng, P. S. and Shokes, F. M. 1991. Disease assessment terms and concepts. *Plant Dis.* 75:1187-1188.
- Pydipati, R., Burks, T. F. and Lee, W. S. 2006. Identification of citrus disease using color texture features and discriminant analysis. *Comput. Electron. Agric.* 52:49-59.
- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., ... Sutskever, I. 2021, July. Learning transferable visual models from natural language supervision. In: Proc. Int. Conf. Mach. Learn., pp. 8748-8763. PMLR, Vienna, Austria.
- Ramcharan, A., Baranowski, K., McCloskey, P., Ahmed, B., Legg, J. and Hughes, D. P. 2017. Deep learning for image-based cassava disease detection. *Front. Plant Sci.* 8:1852.
- Redmon, J., Divvala, S., Girshick, R. and Farhadi, A. 2016. You only look once: Unified, real-time object detection. In: Proc. IEEE Conf. Comput. Vis. Pattern Recognit., pp. 779-788. IEEE, Las Vegas, NV, USA.
- Ren, S., He, K., Girshick, R. and Sun, J. 2017. Faster R-CNN: towards real-time object detection with region proposal networks. *IEEE Trans. Pattern Anal. Mach. Intell.* 39:1137-1149.
- Rumpf, T., Mahlein, A. K., Steiner, U., Oerke, E. C., Dehne, H. W. and Plümer, L. 2010. Early detection and classification of plant diseases with support vector machines based on hyperspectral reflectance. *Comput. Electron. Agric.* 74:91-99.
- Savary, S., Willocquet, L., Pethybridge, S. J., Esker, P., McRoberts, N. and Nelson, A. 2019. The global burden of pathogens and pests on major food crops. *Nat. Ecol. Evol.* 3:430-439.
- Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D. and Batra, D. 2017. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In: Proc. IEEE Int. Conf. Comput. Vis., pp. 618-626. IEEE, Venice, Italy.
- Shafik, W., Tufail, A., De Silva, L. C., Haji Mohd Apong, R. A. A. and Kim, K. H. 2025. Deep learning technique for plant disease classification and pest detection and model explainability elevating agricultural sustainability. *BMC Plant Biol.* 25:1491.
- Simonyan, K. and Zisserman, A. 2014. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*.
- Singh, A. K., Rao, A., Chattopadhyay, P., Maurya, R. and Singh, L. 2024. Effective plant disease diagnosis using vision transformer trained with leafy-generative adversarial network-generated images. *Expert Syst. Appl.* 254:124387.
- Singh, D., Jain, N., Jain, P., Kayal, P., Kumawat, S. and Batra, N. 2020. PlantDoc: A dataset for visual plant disease detection. In: Proc. ACM IKDD CoDS and COMAD, pp. 249-253. ACM, Hyderabad, India.
- Sujatha, R., Krishnan, S., Chatterjee, J. M. and Gandomi, A. H. 2025. Advancing plant leaf disease detection integrating machine learning and deep learning. *Sci. Rep.* 15:11552.
- Tan, M. and Le, Q. 2019. EfficientNet: Rethinking model scaling for convolutional neural networks. In: Proc. Int. Conf. Mach. Learn., pp. 6105-6114. PMLR, Long Beach, CA, USA.
- Thai, H. T., Le, K. H. and Nguyen, N. L. T. 2023. FormerLeaf: An efficient vision transformer for cassava leaf disease detec-

- tion. *Comput. Electron. Agric.* 204:107518.
- Too, E. C., Yujian, L., Njuki, S. and Yingchun, L. 2019. A comparative study of fine-tuning deep learning models for plant disease identification. *Comput. Electron. Agric.* 161:272-279.
- United Nations Department of Economic and Social Affairs. 2024. World population prospects 2024: Summary of results. United Nations, New York, USA.
- United Nations General Assembly. 2015. Transforming our world: the 2030 Agenda for Sustainable Development. URL <https://sdgs.un.org/2030agenda> [9 January 2026].
- Van Horn, G., Mac Aodha, O., Song, Y., Cui, Y., Sun, C., Shepard, A., ... Belongie, S. 2018. The iNaturalist species classification and detection dataset. In: Proc. IEEE Conf. Comput. Vis. Pattern Recognit., pp. 8769-8778. IEEE, Salt Lake City, UT, USA.
- Wang, C. Y., Bochkovskiy, A. and Liao, H. Y. M. 2023. YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. In: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., pp. 7464-7475. IEEE, Vancouver, Canada.
- Wolfert, S., Ge, L., Verdouw, C. and Bogaardt, M. J. 2017. Big data in smart farming – A review. *Agric. Syst.* 153:69-80.
- Zhang, X., Han, L., Dong, Y., Shi, Y., Huang, W., Han, L., et al. 2019. A deep learning-based approach for automated yellow rust disease detection from high-resolution hyperspectral UAV images. *Remote Sens.* 11:1554.